P-W-120C
April 22, 1975
SUPERSEDING
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August 9, 1966

FEDERAL SPECIFICATION

WAX, AUTOMOBILE (PASTE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one grade and two types of automobile paste wax, hereafter called auto wax (see 6.1).

1.2 Classification. The auto wax covered by this specification shall be of the following types as specified (see 6.2):

Type I - Wax with silicone oils.
Type II - Wax (without silicone oils).

2. APPLICABLE DOCUMENTS

2.1 The following documents, is the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28-Gage and Lighter.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 141/6152 - Paint, Varnish, Lacquer, and Related Materials; Methods, Inspection, Sampling, and Testing.
Fed. Test Method 141/6152 - Accelerated Wheathering (Enclosed Arc Apparatus).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials:

D-523 - Specular Gloss.
D-1012 - Aniline Point and Mixed Aniline Point of Hydrocarbon Solvents.
D-1321 - Needle Penetration of Petroleum Waxes.
D-1437 - Flash Point of Solvent-Type Liquid Waxes.
D-2834 - Nonvolatile Matter (Total Solids) in Water Emulsion Waxes, Solvent-Based Waxes and Polymer-Emulsion Floor Waxes.
G-26 - Operating Light-and Water-Exposure Apparatus (Xenon-Arc Type) For Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Auto wax shall consist of natural or synthetic waxes, or a mixture of both, and aliphatic solvents. Type I shall also contain dimethyl siloxane polymer (silicone oil).

3.2 Gloss When tested as specified in 4.4.1, the wax shall produce an increase of 15 gloss units over the unwaxed test panels.

3.3 Weatherability. When tested as specified in 4.4.2, the wax shall retain 65 percent of its initial gloss.

3.4 Stability. When tested as specified in 4.4.3, the wax shall be stable, showing no evidence of separation.

3.5 Chemical and physical properties. The wax, and distilled solvent of the wax, shall conform to the requirements of table I when tested as specified in 4.5.

TABLE I. Requirements

Properties	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Wax:				
Nonvolatile matter (percent)	25.0	-	21.0	31.0
Solvent insoluble matter (percent)	-	0.2	-	0.2
Consistency (cone penetration) tenths of millimeters (mm.)	-	85	-	85
Flashpoint (°C.)	28.0	-	28.0	-
Aniline point (°C.)	46.0	-	46.0	-
Silicone oil (percent)	3.5	10.0	0	-

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance. From each inspection lot offered for Government inspection, four 1-pound containers shall be randomly selected, marked, and forwarded to the laboratory designated to perform acceptance testing.

4.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defects
Containers	Not as specified.
Contents	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers cushioning inadequate, improper, or missing.

4.3.1 Examination of closure, waterproofing, and banding of containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be in accordance with the appendix of that specification.

4.4 Performance characteristics.

4.4.1 Gloss.

4.4.1.1 Preparation of test panels. Obtain steel panels coated with black automobile lacquer 1/, and wash them clean with alcohol followed by a rinse with tap water. Blot dry with a paper towel.

1/ Such 4 x 6" panels are available from the Ford Motor Company, Central Labs Services, 3001 Miller Road, Dearborn, MI 48121.

4.4.1.2 Application of wax. Using a soft cloth, apply a small amount of wax to the test panel, rubbing in a circular motion with the end of one finger. Use moderate pressure for a few seconds to apply a thin, even layer of wax. Allow the waxed panel to dry for 5 to 6 minutes at room temperature and examine visually. Panel should be free from streaking and irregular deposits of polish. Buff lightly with lambs wool for 3 to 4 seconds and measure gloss.

4.4.1.3 Gloss measurement. Gloss measurement shall be carried out with an instrument conforming to ASTM D 523, using an angle of reflectance of 60 degrees. Measure five unwaxed test panels and five waxed test panels, and determine the difference of the averages. This figure should be at least 15 gloss units as per 3.2.

4.4.2 Gloss Retention After Weathering.

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4.4.2.1 Weathering apparatus. Use a weathering apparatus for exposing polished panels to cycles of carbon arc light and water spray as specified in Fed. Test Method Std. No. 141 /6152. Alternately, the same exposure may be produced with a Xenon-Arc apparatus conforming to ASTM G-26.

4.4.2.2 Gloss retention efficiency after weathering . Place six polished panels in an accelerated weathering apparatus as specified in 4.4.2.1 and subject them to a cycle of 102 minutes of carbon arc light and 18 minutes of water spray for a 48 hour period. Remove the panels from the apparatus and wipe them lightly with a soft cloth. Let the panels dry and then determine the gloss of each panel within 24 hours as specified in 4.4.1.3. Determine the gloss retention efficiency of each panel.

Calculation:

$$\text{Gloss retention efficiency} = 1.00 - \frac{(Gp - Gw)}{(Gp - Go)} \times 100$$

Where: Gp = Gloss of waxed unweathered panel.

Gw = Gloss of waxed panel after weathering.

Go = Gloss of unwaxed test panel.

Average the results obtained on five panels to get the average gloss retention efficiency.

4.4.3 Stability.

4.4.3.1 High temperature stability. Place an unopened container of auto wax in an oven maintained at 42.5 ± 0.5°C for 72 hours. After cooling to room temperature, examine visually for any separation of liquid.

4.4.3.2 Freeze-thaw stability. Place an unopened container of auto wax in a freezer chest at -30°C for 6 hours, and allow to thaw overnight at room temperature. Repeat the cycle three times in all, and examine visually for any separation of liquid.

4.5 Chemical and physical properties.

4.5.1 Nonvolatile matter. Nonvolatile matter shall be determined in accordance with ASTM D 2834, the average of two determinations being reported to the nearest 0.1 percent.

4.5.2 Solvent insoluble matter. Weigh 2 to 3 grams (g) wax to the nearest milligram (mg) and dissolve in 100 ml reagent grade trichloroethylene. Examine visually; if the wax is not completely dissolved, filter the solution through a tared Gooch crucible, wash with trichlorethylene, and dry in an oven held at 105 ± 1°C until the crucible reaches constant weight. The percent of insoluble matter is the weight of the residue divided by the weight of the sample multiplied by 100.

4.5.3 Consistency. Determine the consistency of the auto wax as received in the can by measuring the surface face only by the block penetration method in accordance with ASTM D 1321.

4.5.4 Flash point. The flash point shall be determined according to ASTM D 1437.

4.5.5 Aniline point. The aniline point shall be determined according to ASTM D 1012.

4.5.6 Silicone oil. Silicone oil is to be determined quantitatively using infrared spectrophotometry, since silicones used in the type I wax show very strong absorption peaks near 810 cm -1, 1060 cm -1, and 1260 cm -1. A Barnes 0004 series demountable infrared cell or equal may be used with the auto wax.

It is suggested that calibration curves be prepared to facilitate converting absorbance into percent silicone oil. This involves preparing a series of waxes containing known quantities of silicone oil between 0 and 10 percent, and plotting concentration as absorbance on semilog paper.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Sixteen ounces (1 pound) of wax of one type only shall be furnished in a can conforming to PPP-C-96, type V, class 1 or 2, exterior coating plan B.

5.1.1.2 Intermediate packaging. Twelve cans of wax of one type only shall be packaged in a box conforming to PPP-B-566, variety 2, or PPP-B-636, class weather-resistant. The box shall be closed to prevent accidental opening.

5.1.2 Level B.

5.1.2.1 Unit packaging. Sixteen ounces (1 pound) of wax of one type only shall be furnished in a can specified in 5.1.1.1 except the exterior coating shall be plan A of PPP-C-96.

5.1.2.2 Intermediate packaging. Twelve cans of wax shall be intermediately packaged as specified in 5.1.1.2 except the box shall be variety 1 or class domestic.

5.1.3 Level C. The wax shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Cans of wax packaged as specified in 5.1 shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix of the box specification.

5.2.2 Level B. Four intermediate packages specified in 5.1 shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with Method II of the appendix to the box specification.

5.2.3 Level C. The wax in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified (see 6.2).

6. NOTES

6.1 Intended use. The auto waxes covered by this specification are intended for use on reasonably clean automobiles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Packing and packaging required (see section 5).
(d) Quantities required.
(e) Marking document required (see 5.3).

The Defense Department has waived coordination on all revisions and amendments until further notice

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